Problem 7 | Problems T4_PhaseDiagrams I.pdf #7 | Ag-Cu · hypoeutectic

The Hypoeutectic Ag-10 wt% Cu Alloy

Original statement · Problems T4_PhaseDiagrams I.pdf, p. 2

1. *Using the Cu-Ag phase diagram, for an alloy with composition C_0 = 10% Cu, for the temperatures $T_1 = 900$ °C, $T_2 = (780 + \Delta T)$ °C, $T_3 = (780 - \Delta T)$ °C, determine: a. Present phases, chemical composition of phases and mass fraction. b. Microconstituents, chemical composition and mass fraction at T_3.*

Official Solution: a) T_1: Liquid (10%Cu); T_2: C_L = 28.1%Cu; C_α = 8.8%Cu; $X_\alpha = 0.94$; $X_L = 0.06$; T_3: C_α = 8.8%Cu; C_β = 92%Cu; $X_\alpha = 0.98$; $X_\beta = 0.02$. b) eutectic microconstituent ($X_E = 0.06$) and proeutectic α ($X_{\alpha p} = 0.94$)

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Equilibrium cooling**, Ag-Cu diagram of the problem sheet (p. 2): eutectic 28.1 wt% Cu at 779.1 °C ≈ 780 °C; α = (Ag) of 8.8 and β = (Cu) of 92 wt% Cu.
2. **Reading of the figure at 900 °C:** the Ag-side liquidus falls from 961.93 °C (0 % Cu) to 779.1 °C (28.1 % Cu), and at 10 wt% Cu it is just below 900 °C, so at 900 °C the alloy is still liquid (as the official solution states).
3. $8.8 < C_0 = 10 < 28.1$: a **hypoeutectic** alloy, very close to the α limit; the primary phase is α [Session 7 Slide 30].

Data

Temperature	Region	Phase compositions (wt% Cu)
$T_1 = 900$ °C	L	$C_L = 10$
$T_2 = 780 + \Delta T$	α + L	$C_\alpha = 8.8$, $C_L = 28.1$
$T_3 = 780 - \Delta T$	α + β	$C_\alpha = 8.8$, $C_\beta = 92$

PHASE 2 · FRAMES & STRATEGY

Frame: the Ag-Cu diagram (wt% Cu, °C), one tie line per temperature; no change of basis. Lever rule for the phases [Session 7 Slide 16] and for the microconstituents (short tie line from α to the eutectic liquid) [Session 7 Slide 32].

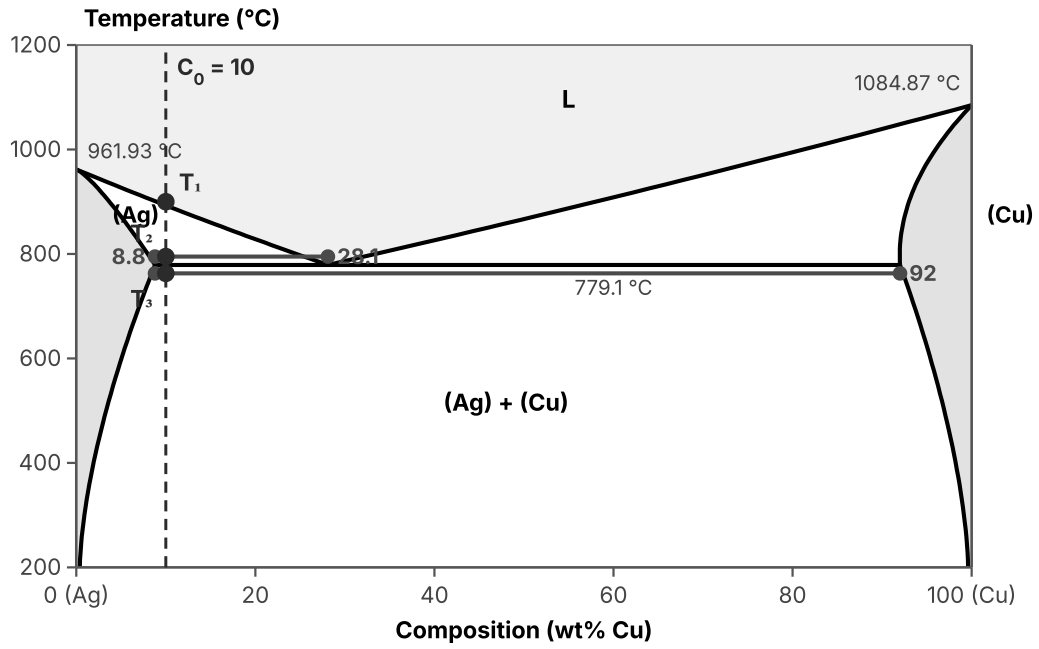


Figure 11. Ag-10 wt% Cu at T_1 = 900 °C (liquid, just above the liquidus, a reading of the figure), T_2 = 780 + ΔT and T_3 = 780 − ΔT. Problem sheet diagram.

PHASE 3 · STEP-BY-STEP DERIVATION

a) Phases $T_1 = 900$ °C. Liquid, $C_L = 10$ wt% Cu, $W_L = 1$.

$T_2 = 780 + \Delta T$ (tie line length $28.1 - 8.8 = 19.3$):

$$W_\alpha = \frac{C_L - C_0}{C_L - C_\alpha} = \frac{28.1 - 10}{28.1 - 8.8} = \frac{18.1}{19.3} = 0.938 \qquad W_L = \frac{C_0 - C_\alpha}{C_L - C_\alpha} = \frac{10 - 8.8}{19.3} = \frac{1.2}{19.3} = 0.062$$

$T_3 = 780 - \Delta T$ (length $92 - 8.8 = 83.2$):

$$W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha} = \frac{92 - 10}{92 - 8.8} = \frac{82}{83.2} = 0.986 \qquad W_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha} = \frac{10 - 8.8}{83.2} = \frac{1.2}{83.2} = 0.014$$

b) Microconstituents at T_3 Proeutectic (primary) α of 8.8 wt% Cu and **eutectic** of 28.1 wt% Cu overall:

$$W_{\alpha p} = \frac{C_e - C_0}{C_e - C_\alpha} = \frac{18.1}{19.3} = 0.938 \qquad W_E = \frac{C_0 - C_\alpha}{C_e - C_\alpha} = \frac{1.2}{19.3} = 0.062$$

Phases inside the eutectic:

$$W_{\alpha,E} = W_\alpha(T_3) - W_{\alpha p} = 0.986 - 0.938 = 0.048 \qquad W_{\beta,E} = W_\beta(T_3) = 0.014$$

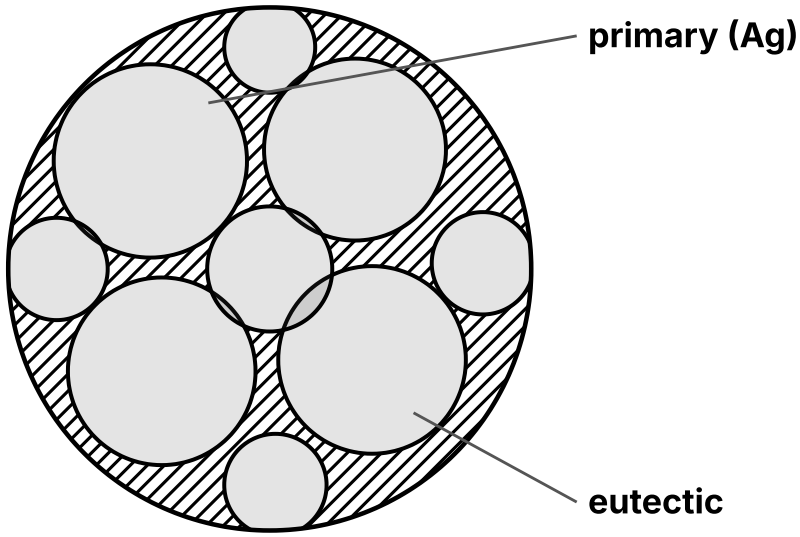


Figure 12. Ag-10 wt% Cu below 779 °C: primary (Ag) (94 %) with a little eutectic (6 %) between the grains.

PHASE 4 · CHECKS & INTERPRETATION

Checks

- $W_{\alpha,E} + W_{\beta,E} = 0.048 + 0.014 = 0.062 = W_E$; inside the eutectic α is $0.048/0.062 = 0.77$, as in Problem 5.
- Mass balance at T_3: $0.9856 \times 8.8 + 0.0144 \times 92 = 8.67 + 1.33 = 10.0$ wt% Cu.

Interpretation The alloy is only 1.2 wt% Cu beyond the solubility limit of α, so almost all of it is primary (Ag) and just 6 % is eutectic, found as thin films between the α grains. An alloy below 8.8 wt% Cu would form no eutectic at all.

Check against the official solution

At T_1, T_2 and in part (b) the results agree with the key.

DISCREPANCY WITH THE OFFICIAL SOLUTION

At T_3 the key gives $X_\alpha = 0.98$ and $X_\beta = 0.02$. With the data of the problem, $W_\alpha = 82/83.2 = 0.9856$ and $W_\beta = 1.2/83.2 = 0.0144$, which round to 0.99 and 0.01. The computed values are kept; the key's 0.98 and 0.02 are not reproduced by any rounding of the data.